

Christ University Kengeri Campus

Smart Map & AI Assistant Navigator

Technical Architecture & Implementation Blueprint

Christ University Kengeri Campus Smart Map

A simple, user-friendly blueprint of the Campus Navigator and Smart Chatbot system.

1. What is this project?

An interactive web application designed to help students, faculty, and visitors easily find classrooms, dining places, hostels, sports amenities, and teacher cabins on the Kengeri campus.

2. Core Features

- Interactive Map: A visual campus map displaying all buildings with color-coded markers (Academic, Dining, Hostels, Sports).
 - Walkway Routes: Instantly draws the shortest walking path between any two locations using Dijkstra's routing algorithm.
 - Indoor Floor Plans: Interactive layouts showing classroom locations on different floors inside campus blocks.
 - Smart Assistant: A floating chat widget that answers questions (like *"Where is Dr. Anjali?"*) and automatically sets routes or locates buildings on the map.
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3. How does it work? (Simple Flow)

1. User asks question: A student clicks a question chip in the chat assistant (e.g., *"Find Cabin"*).
 2. Chatbot processes: The chatbot identifies the faculty member or building.
 3. Map updates: The chatbot sends a message to the main page, which automatically centers the map on the target building and calculates the path.
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4. How to Run the Project

1. Open the project folder.
2. Double-click index.html to launch the map dashboard in your web browser.
3. Click the message icon in the bottom-right corner to open the AI Assistant.